

MAT 324 HW09

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Due 11/13

Problem 1. Let $(X, \mathcal{S}, \mu)$ be a measure space, $p, q, r \in (0, \infty]$, and $f, g: X \rightarrow \mathbb{R}$ be $\mathcal{S}$ -measurable functions.

(a) Suppose $r \leq p, q$ and $1/p + 1/q = 1/r$. Show that $\|fg\|_r \leq \|f\|_p \|g\|_q$.

(b) Suppose $p \leq r \leq q$ and $\alpha, \beta \in [0, 1]$ such that $\alpha + \beta = 1$ and $\alpha/p + \beta/q = 1/r$. Show that

$$\|f\|_r \leq \|f\|_p^\alpha \|f\|_q^\beta.$$

Solution. (a) If $r = \infty$ then $p = q = \infty$ as well, in which case $|f| \leq \|f\|_\infty$ and $|g| \leq \|g\|_\infty$ almost everywhere, so $|fg| \leq \|f\|_\infty \|g\|_\infty$ almost everywhere, which implies $\|fg\|_\infty \leq \|f\|_\infty \|g\|_\infty$. Otherwise, if r is finite, apply Hölder's inequality on the functions $|f|^r, |g|^r$ with exponents p/r and q/r to get

$$\|fg\|_r = \| |f|^r |g|^r \|_1^{1/r} \leq \| |f|^r \|_{p/r}^{1/r} \| |g|^r \|_{q/r}^{1/r} = \|f\|_p \|g\|_q.$$

(b) If $r = \infty$ then $p = q = \infty$ as well, so the inequality is trivial. Also, if $\alpha = 0$ or $\beta = 0$, then respectively $q = r$ or $p = r$, and in each case the inequality is trivial. Suppose now that $\alpha, \beta \in (0, 1)$ and r is finite; then, apply Hölder's inequality on the functions $|f|^{\alpha r}, |f|^{\beta r}$ with exponents $p/(\alpha r)$ and $q/(\beta r)$ to get

$$\|f\|_r = \| |f|^r \|_1^{1/r} \leq \| |f|^{\alpha r} \|_{p/(\alpha r)}^{1/r} \| |f|^{\beta r} \|_{q/(\beta r)}^{1/r} = \|f\|_p^\alpha \|f\|_q^\beta. \quad \square$$

Problem 2. Find a continuous function $f: \mathbb{R}^+ \rightarrow \mathbb{R}^+$ which is in $\mathcal{L}^2(\mathbb{R}^+)$, but not in $\mathcal{L}^p(\mathbb{R}^+)$ for any $p \in (0, \infty] - \{2\}$. Justify your answer.

Solution. Let $f(x) = x^{-1/2}(\ln(x)^2 + 1)^{-1}$. We can bound

$$\begin{aligned} \int_{(0,1/2)} f^2 d\lambda &\leq \int_{(0,1/2)} \frac{1}{x \ln(x)^4} d\lambda(x) = \lim_{a \rightarrow 0} \frac{1}{3} (\ln(a)^{-3} - \ln(1/2)^{-3}) = -\frac{\ln(1/2)^{-3}}{3} < \infty, \\ \int_{(2,\infty)} f^2 d\lambda &\leq \int_{(0,1/2)} \frac{1}{x \ln(x)^4} d\lambda(x) = \lim_{b \rightarrow \infty} \frac{1}{3} (\ln(2)^{-3} - \ln(b)^{-3}) = \frac{\ln(2)^{-3}}{3} < \infty, \end{aligned}$$

and f is bounded over $[1/2, 2]$, so the sum of the integrals of f over $(0, 1/2)$, $[1/2, 2]$, and $(2, \infty)$, which equals the integral of f over $\mathbb{R}^+$, is finite; thus, $f \in \mathcal{L}^2(\mathbb{R}^+)$.

Let $p < 2$. For any $\varepsilon > 0$, there exists $C_\varepsilon > 0$ such that $x > C_\varepsilon$ implies $\ln(x)^2 + 1 < x^\varepsilon$, so if $\varepsilon > 0$ is such that $p/2 + \varepsilon < 1$, then $x > C_\varepsilon$ implies $f^p \geq x^{-p/2-\varepsilon}$, meaning

$$\int_{(0,\infty)} f^p d\lambda \geq \int_{(C_\varepsilon,\infty)} \frac{1}{x^{p/2+\varepsilon}} d\lambda(x) \geq \lim_{b \rightarrow \infty} \frac{1}{1 - p/2 - \varepsilon} (b^{-p/2-\varepsilon+1} - C_\varepsilon^{-p/2-\varepsilon+1}) = \infty,$$

hence f is not in $\mathcal{L}^p$ in this case.

Let $p > 2$. For any $\varepsilon > 0$, there exists $c_\varepsilon > 0$ such that $x < c_\varepsilon$ implies $\ln(x)^2 + 1 < x^{-\varepsilon}$, so if $\varepsilon > 0$ is such that $p/2 - \varepsilon > 1$, then $x < c_\varepsilon$ implies $f^p \geq x^{-p/2+\varepsilon}$, meaning

$$\int_{(0,\infty)} f^p d\lambda \geq \int_{(0,c_\varepsilon)} \frac{1}{x^{p/2-\varepsilon}} d\lambda(x) \geq \lim_{a \rightarrow 0} \frac{1}{p/2 - \varepsilon - 1} (a^{-p/2+\varepsilon+1} - c_\varepsilon^{-p/2+\varepsilon+1}) = \infty,$$

hence f is not in $\mathcal{L}^p$ in this case. $\square$

Problem 3. For each of the following sequences of measurable functions $f_1, f_2, \dots : \mathbb{R}^+ \rightarrow \mathbb{R}$, determine whether the sequence lies in $L^1(\mathbb{R}^+)$, $L^2(\mathbb{R}^+)$, and if so, whether it is Cauchy in there. Justify your answers.

$$(a) f_n = \mathbb{1}_{(0,n)}/\sqrt{x}, \quad (b) f_n = \mathbb{1}_{(0,n)}/(x+1).$$

Solution. (a) $f_n \in L^1(\mathbb{R}^+)$ because

$$\int_{(0,n)} |f_n| \, d\lambda = \lim_{c \rightarrow 0} \int_c^n \frac{1}{\sqrt{x}} \, dx = \lim_{c \rightarrow 0} \frac{2}{3} (n\sqrt{n} - c\sqrt{c}) = \frac{2n\sqrt{n}}{3} < \infty.$$

However, the sequence is not Cauchy in $L^1(\mathbb{R}^+)$ because for any $m, n \in \mathbb{N}$ with $m > n$,

$$\|f_m - f_n\|_1 = \int_n^m \frac{1}{\sqrt{x}} \, dx = \frac{2(m\sqrt{m} - n\sqrt{n})}{3},$$

which is unbounded as m grows arbitrarily large.

Additionally, $f_n \notin L^2(\mathbb{R}^+)$ because for all $n \in \mathbb{N}$,

$$\int_{(0,n)} f_n^2 \, d\lambda = \lim_{c \rightarrow 0} \int_c^n \frac{1}{x} \, dx = \lim_{c \rightarrow 0} (\ln n - \ln c) = \infty$$

(b) $f_n \in L^1(\mathbb{R}^+)$ because

$$\|f_n\|_1 = \int_0^n \frac{1}{x+1} \, d\lambda = \ln(n+1) - \ln(1) = \ln(n+1) < \infty,$$

but f_n is not Cauchy in $L^1(\mathbb{R}^+)$ because for any $m, n \in \mathbb{N}$ with $m > n$,

$$\|f_m - f_n\|_1 = \int_n^m \frac{1}{x+1} \, dx = \ln(m+1) - \ln(n+1),$$

which is unbounded as m grows arbitrarily large.

Additionally, $f_n \in L^2(\mathbb{R}^+)$ because $0 \leq f_n^2 < f_n$ for every n , so the integral of f_n^2 is finite because the integral of f_n is finite. Moreover, the sequence is Cauchy in $L^2(\mathbb{R}^+)$ because given $\varepsilon > 0$, if N is such that $1/\sqrt{N} < \varepsilon$, then for integers $m > n > N$,

$$\|f_m - f_n\|_2 = \left(\int_n^m \frac{1}{(x+1)^2} \right)^{1/2} = \left(\frac{1}{n} - \frac{1}{m} \right)^{1/2} < \sqrt{\frac{1}{n}} < \sqrt{\frac{1}{N}} < \varepsilon. \quad \square$$

Problem 4 (Axler 7B #12). Show that there exists a sequence of functions $f_1, f_2, \dots \in \mathcal{L}^1([0, 1])$ such that $\lim_{k \rightarrow \infty} \|f_k\|_1 = 0$ but

$$\sup\{f_k(x) : k \in \mathbb{N}\} = \infty$$

for every $x \in [0, 1]$.

Solution. Given $k \in \mathbb{N}$, let $m, r \in \mathbb{Z}_{\geq 0}$ be such that $k = 2^m + r$ with r as small as possible, and define $f_k = m \cdot \mathbb{1}_{[r/2^m, (r+1)/2^m]}$, so for example $f_1 \equiv 0$ and $f_{10} = 3 \cdot \mathbb{1}_{[2/8, 3/8]}$. Then, for each $k \in \mathbb{N}$,

$$\|f_k\|_1 = \int_{[0,1]} |f_k| \, d\lambda = \int_{[0,1]} m \cdot \mathbb{1}_{[r/2^m, (r+1)/2^m]} \, d\lambda = \frac{m}{2^m},$$

which becomes arbitrarily small as k , and thus m , goes to infinity. On the other hand, for every $m \in \mathbb{Z}_{\geq 0}$,

$$\bigcup_{r=0}^{2^m-1} [r/2^m, (r+1)/2^m] = [0, 1]$$

so given $x \in [0, 1]$ and $m \in \mathbb{Z}_{\geq 0}$, there exists a k such that $f_k(x) = m$, hence $\sup\{f_k(x) : k \in \mathbb{N}\} = \infty$. $\square$

Problem 5. Let $p, q \in [1, \infty]$ with $1/p + 1/q = 1$. For a differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$, define

$$\|f\|_{p,1} = \|f\|_p + \|f'\|_p.$$

(a) Show that this defines a norm on the vector space

$$C^{1,p}(\mathbb{R}) := \{f \in C^1(\mathbb{R}) : \|f\|_{p,1} < \infty\}.$$

Do not forget to justify why $C^{1,p}(\mathbb{R})$ is a vector space.

(b) Show that

$$|f(x) - f(y)| \leq \|f'\|_p |x - y|^{1/q}, \quad \left| f(x) - \frac{1}{2} \int_{x-1}^{x+1} f(t) dt \right| \leq \|f'\|_p, \quad \|f\|_\infty \leq \|f\|_{p,1}$$

for all $f \in C^{1,p}(\mathbb{R})$ and $x, y \in \mathbb{R}$.

(c) Let $f_1, f_2, \dots \in C^{1,p}(\mathbb{R})$ be a Cauchy sequence with respect to the norm $\|\bullet\|_{p,1}$. Show that it converges uniformly to a bounded continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$.

Solution. (a) Suppose $f, g: \mathbb{R} \rightarrow \mathbb{R} \in C^{1,p}(\mathbb{R})$ and $c \in \mathbb{R}$. Since $p \geq 1$, Minkowski's inequality gives

$$\begin{aligned} \|f + g\|_{p,1} &= \|f + g\|_p + \|f' + g'\|_p \leq \|f\|_p + \|g\|_p + \|f'\|_p + \|g'\|_p = \|f\|_{p,1} + \|g\|_{p,1} < \infty, \\ \|cf\|_{p,1} &= \|cf\|_p + \|cf'\|_p = |c| \|f\|_p + |c| \|f'\|_p = |c| \|f\|_{p,1} < \infty. \end{aligned}$$

This proves both that $f + g$ and cf are in $C^{1,p}(\mathbb{R})$, making it a vector space, and that $\|\bullet\|_{p,1}$ satisfies two of the properties to be a norm. Moreover, if $\|f\|_{p,1} = 0$, then $\|f\|_p = 0$, so $f = 0$ almost everywhere; that is, $f^{-1}(\mathbb{R} \setminus \{0\})$ has measure zero. By continuity, this set is an open subset of $\mathbb{R}$, so it must be empty because nonempty open subsets of $\mathbb{R}$ have positive measure, i.e. $f \equiv 0$. Thus, $\|\bullet\|_{p,1}$ is a norm on $C^{1,p}(\mathbb{R})$.

(b) Suppose that $x \leq y$ and apply Hölder's inequality on the restriction of the Lebesgue measure to $[x, y]$ with functions f' , 1 and exponents p, q to get via the fundamental theorem of calculus that

$$\begin{aligned} |f(x) - f(y)| &= \left| \int_x^y f'(t) dt \right| \leq \int_x^y |f'(t)| dt \\ &\stackrel{\text{Hölder}}{\leq} \|f'\|_{[x,y]} \|1\|_{[x,y]} \leq \|f'\|_p \lambda([x,y])^{1/q} = \|f'\|_p |x - y|^{1/q}. \end{aligned} \quad (1)$$

Then, $\frac{1}{2} \int_{x-1}^{x+1} f(t) dt = f(\xi)$ for some $\xi \in [x-1, x+1]$ by the mean value theorem on $F(u) = \int_{-\infty}^u f(t) dt$. In particular $|x - \xi| \leq 1$, so by the previous inequality,

$$\left| f(x) - \frac{1}{2} \int_{x-1}^{x+1} f(t) dt \right| = |f(x) - f(\xi)| \stackrel{(1)}{\leq} \|f'\|_p |x - \xi|^{1/q} \leq \|f'\|_p. \quad (2)$$

Finally, for each $x \in \mathbb{R}$,

$$\begin{aligned} |f(x)| &\leq \left| f(x) - \frac{1}{2} \int_{x-1}^{x+1} f(t) dt \right| + \left| \frac{1}{2} \int_{x-1}^{x+1} f(t) dt \right| \\ &\stackrel{\text{Hölder}, (2)}{\leq} \|f'\|_p + \frac{1}{2} \|f\|_{[x-1, x+1]} \|1\|_{[x-1, x+1]} \leq \|f'\|_p + \|f\|_p \cdot 2^{1/q-1} \leq \|f'\|_p + \|f\|_p = \|f\|_{p,1}, \end{aligned}$$

noting that $1/q - 1 \leq 0$ because $q \geq 1$; taking the supremum over $x \in \mathbb{R}$ yields $\|f\|_\infty \leq \sup_{\mathbb{R}} |f| \leq \|f\|_{p,1}$.

(c) First, note that if $g \in C^{1,p}(\mathbb{R})$ then $|g|^p$, hence $|g|$, is bounded; also, $\|g\|_\infty = \sup_{\mathbb{R}} |g|$ because if $M = \sup_{\mathbb{R}} |g|$ then for any $\varepsilon > 0$, the set $g^{-1}((M - \varepsilon, \infty))$ has positive measure (being nonempty and open in $\mathbb{R}$), meaning one cannot modify g on a set of measure zero to reduce the upper bound.

Given $\varepsilon > 0$, there exists $N_\varepsilon \in \mathbb{N}$ such that $m, n \geq N_\varepsilon$ implies $\|f_m - f_n\|_\infty \leq \|f_m - f_n\|_{p,1} < \varepsilon$; thus, f_n is Cauchy with respect to $\|\bullet\|_\infty$ (i.e. the sup norm) and that $\|f_n\|_\infty$ is a Cauchy sequence. The first statement tells us that f_n is a uniformly converging sequence of continuous functions, so its limit f is continuous; the second tells us that the f_n are uniformly bounded, hence f is bounded. $\square$